

Stochastic systems via rough path theory: theory and numerics

Luu Hoang Duc

(IMH-VAST & MPI MIS)

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Outline

Rough path theory

- Rough differential equations (RDEs)

Discrete rough paths

- Discrete settings

- Discrete sewing lemma

- Rough difference equations

- Numerical errors

Numerical attractors

- Existence of attractors

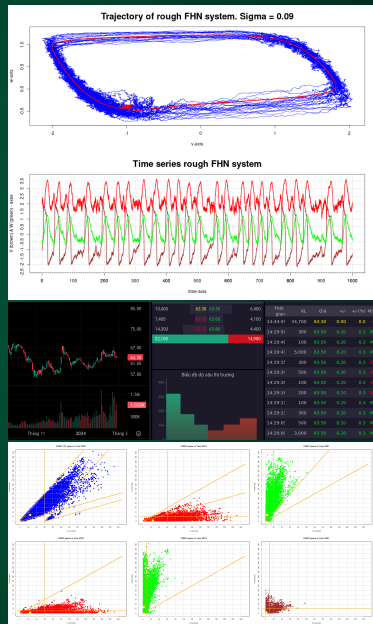
- Semi-continuity of attractors

Applications

Research interests

- Stochastic processes, random dynamical systems, rough path theory, neural (controlled) differential equations
- Learning the stable dynamics with non-Lipschitz and chaotic behaviors: numerical attractors
- Learning the non-autonomous dynamics: trend detection.
- Signature theory for data streams: order book data (finance)
- Topological space of chemical formulae (chemistry).

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- LHD. Discrete & Cont. Dyn. Syst., 2022
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Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^\beta([a, b], \mathbb{R}^d)$ and $\mathbb{X} \in C_2^{2\beta}([a, b]^2, \mathbb{R}^d \otimes \mathbb{R}^d)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad \forall a \leq s \leq u \leq t \leq b. \quad (1.1)$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$. In a general form, $(y \otimes x)_{s,t} =: \int_s^t y_{s,u} \otimes dx_u$ satisfies Chen's relation

$$(y \otimes x)_{s,t} - (y \otimes x)_{s,u} - (y \otimes x)_{u,t} = y_{s,u} \otimes x_{u,t}.$$

Denote by $\mathcal{C}^\beta([a, b]) \subset C^\beta \oplus C^{2\beta}$ the set of all rough paths in $[a, b]$, then $\mathcal{C}^\beta$ is a closed set but not a linear space, equipped with the rough path semi-norm

$$\|\mathbf{x}\|_{\beta, [a, b]} := \|x\|_{\beta, [a, b]} + \|\mathbb{X}\|_{2\beta, [a, b]^2}^{\frac{1}{2}} < \infty. \quad (1.2)$$

Denote by $T^2(\mathbb{R}^d) = \mathbb{R} \oplus \mathbb{R}^d \oplus \mathbb{R}^{d \otimes 2}$ the truncated step-2 tensor algebra, then $T^2(\mathbb{R}^d)$ is a Banach space with the norm $|\omega| := \max_{k=0,1,2} \|\pi_k(\omega)\|$ where $\|\pi_k(\omega)\|$ is Euclidean norm on $(\mathbb{R}^d)^{\otimes k}$. Define $T_1^2(\mathbb{R}^d) = \{\omega \in T^2(\mathbb{R}^d) : \pi_0(\omega) = 1\}$ together with a $\bullet$ operator

$$(1, g^1, g^2) \bullet (1, h^1, h^2) = (1, g^1 + h^1, g^2 + h^2 + g^1 \otimes h^1), \quad (1.3)$$

for any $\mathbf{g} = (1, g^1, g^2)$, $\mathbf{h} = (1, h^1, h^2)$. Then $(T_1^2(\mathbb{R}^d), \bullet)$ is a group with the identity element $\mathbf{1} = (1, 0, 0)$, and each element $\mathbf{g}$ has its inverse element $\mathbf{g}^{-1} = (1, -g^1, -g^2 - g^1 \otimes g^1)$. In addition, for a rough path $\mathbf{x}$ one can easily check from Chen's relation (2.1) that

$$(1, \mathbf{x}_{s,u}, \mathbb{X}_{s,u}) \bullet (1, \mathbf{x}_{u,t}, \mathbb{X}_{u,t}) = (1, \mathbf{x}_{s,t}, \mathbb{X}_{s,t}), \quad \forall s \leq u \leq t \in \Pi.$$

Also, one can view $\mathbf{x}_t = (1, \mathbf{x}_{0,t}, \mathbb{X}_{0,t})$ as a map from Π to $T_1^2(\mathbb{R}^d)$.

$\beta > \frac{1}{2}$: Young integrals

- Consider $y \in C^\beta([a, b], V \otimes H)$ and $x \in C^\beta([a, b], H)$, there exists a Young integral $\int_a^b y dx$ defined by

$$\int_a^b y dx = \lim_{|\Pi| \rightarrow 0} \sum_{[s, t] \in \Pi} y_s x_{s, t},$$

- Young-Loeve (Y-L) estimate

$$\left\| \int_s^t y_u dx_u - y_s x_{s, t} \right\| \leq K_\beta |t - s|^{2\beta} \|y\|_{\beta, [a, b]} \|x\|_{\beta, [a, b]}.$$

In fact, we can prove using fractional calculus that

$$\left\| \int_s^t y_u dx_u - y_s x_{s, t} \right\| \leq K_\beta \left(\|y_{s, \cdot}\|_{\infty, [s, t]} + \|y\|_{\beta, \beta, [s, t]} \right) \left(\|x_{\cdot, t}\|_{\infty, [s, t]} + \|x\|_{\beta, \beta, [s, t_-]} \right).$$

$\beta \in (\frac{1}{3}, \frac{1}{2})$: Rough integrals

- A path $y \in C^\beta([a, b], V \otimes H)$ is then called to be *controlled by* $x \in C^\beta([a, b], H)$ if there exists a tube (y', R^y) with $y' \in C^\beta([a, b], V \otimes H \otimes H)$, $R^y \in C^{2\beta}(\Delta^2([a, b]), V \otimes H)$ such that

$$y_{s,t} = y'_s x_{s,t} + R^y_{s,t}, \quad \forall \min I \leq s \leq t \leq \max I.$$

y' is called **Gubinelli derivative** of y , which is **uniquely defined** as long as $x \in C^\beta \setminus C^{2\beta}$ is **truly rough** (works for fBm B^H , $H \in (\frac{1}{3}, \frac{1}{2}]$ FH[Chapter 6]).

- Denote $F_{s,t} := y_s x_{s,t} + y'_s \mathbb{X}_{s,t}$. Then

$$\delta F_{s,u,t} := F_{s,t} - F_{s,u} - F_{u,t} = -y'_{s,u} \mathbb{X}_{u,t} - R^y_{s,u} x_{u,t}$$

$$\Rightarrow \|\delta F_{s,u,t}\| \leq |t-s|^{3\beta} \left(\|x\|_{\beta,[s,t]} \|R^y\|_{2\beta,\Delta^2[s,t]} + \|y'\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]} \right)$$

Thanks to the sewing lemma, the rough integral $\int_s^t y_u dx_u$ can be defined as

$$\int_s^t y_u dx_u := \lim_{|\Pi| \rightarrow 0} \sum_{[u,v] \in \Pi} [y_u x_{u,v} + y'_u \mathbb{X}_{u,v}]$$

Moreover, one gets Y-L typed estimate:

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} - y'_s \mathbb{X}_{s,t} \right\| \leq C_\beta |t-s|^{3\beta} \left(\|x\|_{\beta,[s,t]} \|R^y\|_{2\beta,\Delta^2[s,t]} + \|y'\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]} \right)$$

Existence and uniqueness theorem: RDEs

- Finite dimension: RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ is understood in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)dx_s. \quad (1.4)$$

- Riedel and Scheutzow (2017): solve the solution in Friz-Victoir sense.
- LHD (2021-2022): solve in Gubinelli sense, i.e. solution of the form (y, y') with $y, y' \in C^\beta$, $y' = g(y)$.
- Wong-Zakai approximation (Ling-Riedel-Scheutzow (2022), Cao-Gao-Schmalfuss (2023))

$$dy_t^\delta = f(y_t^\delta)dt + g(y_t^\delta)dx_t^\delta, \quad (1.5)$$

where either x^δ is a dyadic approximation

$$x_t^\delta = x_{k\delta} + \frac{t - k\delta}{\delta} x_{k\delta, (k+1)\delta}, \quad \forall t \in [k\delta, (k+1)\delta];$$

or δ -shift approximation

$$x_t^\delta := \int_0^t \frac{x_{s, s+\delta}}{\delta} ds$$

Then one can prove that rough solution in Gubinelli sense coincides is also the solution in Friz-Victoir sense.

Finite dimension case: Assumptions

- $(\mathbf{H}_f)$ $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is locally Lipschitz continuous and of one-sided linear growth

$$\exists C > 0 : \langle y, f(y) \rangle \leq C(1 + \|y\|^2), \quad \forall y \in \mathbb{R}^d; \quad (1.6)$$

in addition f is of linear growth in the perpendicular direction, i.e. there exists $C_f > 0$ such that

$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y \neq 0; \quad (1.7)$$

- $(\mathbf{H}_g^b)$ g belongs to $C_b^3(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^m, \mathbb{R}^d))$ such that

$$C_g := \max \left\{ \|g\|_\infty, \|Dg\|_\infty, \|D_g^2\|_\infty, \|D_g^3\|_\infty \right\} < \infty; \quad \text{or} \quad (1.8)$$

- **Existence-Uniqueness theorem:** For each $\lambda \in (0, 1)$ small enough, there exist some generic constants $C(\lambda), \delta(\lambda)$ such that the solution satisfies the following estimates

$$\|y\|_{\infty, [0, T]} \leq e^{\delta(\lambda)T} \left(\|y_0\| + C(\lambda)T + \frac{\lambda}{2} N\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [0, T]\right) \right) =: R.$$

Doss-Sussmann transformation

- Condition (1.6) is equivalent to the following condition

$$\exists \bar{C} > 0 : \langle y, f(y) \rangle \leq \bar{C} \|y\| (1 + \|y\|), \quad \forall y \in \mathbb{R}^d; \quad (1.9)$$

- LHD (2020): the solution $\phi \cdot (\mathbf{x}, \phi_a)$ of $d\phi_u = g(\phi_u)dx_u$, $u \in [a, b]$, $\phi_a \in \mathbb{R}^d$ is C^1 w.r.t. ϕ_s , and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_s))\xi_u dx_u, \quad u \in [a, b], \xi_a = Id, \quad (1.10)$$

- Cass-Litterer-Lyons (2013): **Greedy times**

$\tau_0 = \min I$, $\tau_{i+1} := \inf \left\{ t > \tau_i : \|\mathbf{x}\|_{p, [\tau_i, t]} = \gamma \right\} \wedge \max I$. Assign $N(\gamma, \mathbf{x}, I) := \sup \{i \in \mathbb{N} : \tau_i \leq \max I\}$.

- The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$ there is an one-to-one correspondence between a solution y_t of (2.22) on a certain interval $[0, \tau]$ and a solution z_t of the associate ordinary differential equation

$$\dot{z}_t = \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} f(\phi_t(\mathbf{x}, z_t)), \quad t \in [0, \tau], \quad z_0 = y_0. \quad (1.11)$$

To estimate the solution norm growth, assign

$$\gamma_t := y_t - z_t, \quad \text{and} \quad \psi_t := \left[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t) \right]^{-1} - Id, \quad \forall t \in [0, \tau],$$

Discrete functions and norms

Davie (2008), Lejay (2010) on Euler scheme for RDE.

Bayer-Friz-Tapia (2023): deep neural network stability via discrete rough paths.

Let $[a, b] \subset \mathbb{R}$ and $\Pi = \{a = t_0 < t_1 < \dots < t_n = b\}$ be an arbitrary (finite) set of points of $[a, b]$. We introduce various norms which are natural discrete versions of the continuous ones

$$\|y\|_{\infty, \Pi} := \sup_{t_i \in \Pi} \|y_{t_i}\|;$$

$$\|y\|_{p, \Pi} := \sup_{t_i^* \in \Pi, 0 \leq i \leq r, t_0^* < t_1^* \dots < t_r^*} \left(\sum_{i=0}^{r-1} \|y_{t_i^*} - y_{t_{i+1}^*}\|^p \right)^{1/p};$$

$$\|y\|_{p, \Pi} = \|y_a\| + \|y\|_{p, \Pi};$$

$$\|F\|_{p, \Pi} := \sup_{t_i^* \in \Pi, 0 \leq i \leq r, t_0^* < t_1^* \dots < t_r^*} \left(\sum_{i=0}^{r-1} |F_{t_i^*, t_{i+1}^*}|^p \right)^{1/p},$$

$$\|F\|_{\infty, \Pi} := \sup_{a \leq s \leq t \leq b} \|F_{s, t}\|.$$

For a path ξ we write $\xi_{s, t} = \xi_t - \xi_s$.

Discrete rough paths

- A non negative function ω defined on $\Delta\Pi := \{(s, t) \in \Pi^2 | s \leq t\}$ is called a *discrete control function* on Π if it vanishes on the diagonal, i.e. $\omega_{s,s} = 0, \forall s \in \Pi$, and is superadditive, i.e. for all $s \leq u \leq t$ in Π

$$\omega_{s,u} + \omega_{u,t} \leq \omega_{s,t}.$$

- For given $[a, b]$ and Π a (finite) discrete time set in $[a, b]$ as above, let $x : \Pi \rightarrow \mathbb{R}^m$ and $\mathbb{X}(\cdot, \cdot) : \Pi^2 \rightarrow \mathbb{R}^{m \times m}$ be discrete functions defined on Π and Π^2 respectively. One then says that x can be lifted to a discrete rough path $\mathbf{x} = (x, \mathbb{X})$ if it satisfies Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad s \leq u \leq t; \quad s, u, t \in \Pi. \quad (2.1)$$

One can then furnish $\mathbf{x}$ with a semi-norm

$$\|\mathbf{x}\|_{p, \Pi[a,b]} = \left(\|x\|_{p, \Pi[a,b]}^p + \|\mathbb{X}\|_{p/2, \Pi[a,b]}^{p/2} \right)^{1/p}. \quad (2.2)$$

- Example:

$$\mathbb{X}_{t_k, t_l}^{\mu, \nu} := \sum_{j=k}^{l-1} x_{t_k, t_j}^{(\mu)} x_{t_j, t_{j+1}}^{(\nu)}.$$

Discrete sewing lemma

Let $\Pi = \{t_i : 0 \leq i \leq n, a = t_0 < t_1 < \dots < t_n = b\}$ be a finite partition of $[a, b]$ and F be an function defined on $\Delta\Pi$, vanished on the diagonal, i.e. $F_{s,s} = 0, \forall s \in \Pi$. Put

$$\begin{aligned} (\delta F)_{s,u,t} &:= F_{s,t} - F_{s,u} - F_{u,t}, \quad s \leq u \leq t \in \Pi, \\ I_{k,l} &:= \sum_{k \leq j \leq l-1} F_{t_j, t_{j+1}} - F_{t_k, t_l}, \quad t_k \leq t_l \in \Pi. \end{aligned}$$

Assume that for a discrete control ω on Π and a number $\lambda > 1$ we have for all $s \leq u \leq t$ in Π the following inequality

$$\|(\delta F)_{s,u,t}\| \leq \omega_{s,t}^\lambda. \quad (2.3)$$

Then there exists a constant $K > 0$ depending only on λ such that

$$\|I_{k,l}\| \leq K \omega_{t_k, t_l}^\lambda, \quad \forall t_k \leq t_l \in \Pi. \quad (2.4)$$

Proof of discrete sewing lemma

Fix a $K \geq \frac{2}{1-2^{1-\lambda}} > 2$. We are going to prove (2.4) by induction for $l - k$. It is easy to check that (2.4) holds trivially for $l - k = 0, 1$. Assume it still holds for all k, l such that $k \leq l < m$. We will prove that (2.4) is true for $l = m$. Define

$$l := \max\{k \leq i \leq m : \omega_{t_k, t_i} \leq \frac{1}{2} \omega_{t_k, t_m}\}.$$

By definition, $\omega_{t_k, t_l} \leq \frac{1}{2} \omega_{t_k, t_m}$ and $\omega_{t_k, t_{l+1}} > \frac{1}{2} \omega_{t_k, t_m}$. It follows that $\omega_{t_{l+1}, t_m} < \frac{1}{2} \omega_{t_k, t_m}$ due to super additivity of ω . Since $l_{l, l+1} = 0$, it follows that

$$\begin{aligned} l_{k, m} &= l_{k, l} + l_{l, m} - (\delta F)_{t_k, t_l, t_m} \\ &= l_{k, l} + l_{l, l+1} + l_{l+1, m} - (\delta F)_{t_l, t_{l+1}, t_m} - (\delta F)_{t_k, t_l, t_m} \\ &= l_{k, l} + l_{l+1, m} - (\delta F)_{t_l, t_{l+1}, t_m} - (\delta F)_{t_k, t_l, t_m}. \end{aligned}$$

By inductive hypothesis and assumption of δF we obtain

$$\begin{aligned}
\|I_{k,m}\| &\leq \|I_{k,l}\| + \|I_{l+1,m}\| + \|(\delta F)_{t_l, t_{l+1}, t_m}\| + \|(\delta F)_{t_k, t_l, t_m}\| \\
&\leq K\omega_{t_k, t_l}^\lambda + K\omega_{t_{l+1}, t_m}^\lambda + \omega_{t_l, t_m}^\lambda + \omega_{t_k, t_m}^\lambda \\
&\leq 2^{1-\lambda} K\omega_{t_k, t_m}^\lambda + 2\omega_{t_k, t_m}^\lambda \\
&\leq (2^{1-\lambda} K + 2)\omega_{t_k, t_m}^\lambda \\
&\leq K\omega_{t_k, t_m}^\lambda.
\end{aligned}$$

This completes the proof.

Corollary: Let $\Pi = \{t_i : 0 \leq i \leq n, a = t_0 < t_1 < \dots < t_n = b\}$ be a finite partition of $[a, b]$. Consider a discrete system defined on Π

$$y_{t_{j+1}} = y_{t_j} + F_{t_j, t_{j+1}} + \epsilon_{t_j, t_{j+1}}, \quad y_{t_0} = y^* \in \mathbb{R}^d, \quad j = 0, 1, \dots, n-1, \quad (2.5)$$

where $F : \Pi^2 \rightarrow \mathbb{R}^d$, $\epsilon : \Pi^2 \rightarrow \mathbb{R}^d$. Assume that

- (i) There exists a discrete control ω such that (2.3) is satisfied for F ;
- (ii) There exists a discrete control function $\omega^{(0)}$ such that $|\epsilon_{t_j, t_{j+1}}| \leq \omega_{t_j, t_{j+1}}^{(0)}$.

Then there exists a constant $K > 0$ such that for any $r \geq 1$

$$\|y\|_{r, \Pi[a, b]} \leq K\omega_{a, b}^\lambda + \|F\|_{r, \Pi[a, b]} + \omega_{a, b}^{(0)}. \quad (2.6)$$

Greedy sequence of times

Continuous time version: see Cass-Litterer-Lyons, 2013.

Given a fixed $\gamma > 0$, assign the starting time $\tau_0 = a$. For each $m \in \mathbb{N}$, assume $\tau_m = t_k$ is determined. Then one can define τ_{m+1} by the following rule:

- if $\sum_{\omega \in \mathcal{S}} \omega_{t_k, t_{k+1}}^{\beta_\omega} > \gamma$ then set $\tau_{m+1} := t_{k+1}$;
- else set $\tau_{m+1} := \sup\{t_l \in (t_k, b] : \sum_{\omega \in \mathcal{S}} \omega_{t_k, t_l}^{\beta_\omega} \leq \gamma\}$.

Define $N(\gamma, [a, b])$ to be the number of times τ_m in $[a, b]$. From the definition,

$$\sum_{\omega \in \mathcal{S}} \omega_{\tau_m, \tau_{m+2}}^{\beta_\omega} > \gamma.$$

By taking both sides to the power of $\frac{1}{\beta}$ where $\beta := \min\{\beta_\omega : \omega \in \mathcal{S}\} \leq 1$, and using the Jensen's inequality, one obtains

$$\gamma^{\frac{1}{\beta}} < \left(\sum_{\omega \in \mathcal{S}} \omega_{\tau_m, \tau_{m+2}}^{\beta_\omega} \right)^{\frac{1}{\beta}} \leq (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \left(\sum_{\omega \in \mathcal{S}} \omega_{\tau_m, \tau_{m+2}}^{\frac{\beta_\omega}{\beta}} \right). \quad (2.7)$$

As a result, taking the sum of both sides of (2.7) as m going from 0 to $N(\gamma, [a, b]) - 3$ (whenever $N(\gamma, [a, b]) \geq 3$) and using the fact that all elements in the square brackets of (2.7) are controls, one obtains

$$\begin{aligned} [N(\gamma, [a, b]) - 2]\gamma^{\frac{1}{\beta}} &< \sum_{m=0}^{N(\gamma, [a, b]) - 3} (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \left(\sum_{\omega \in \mathcal{S}} \omega^{\frac{\beta \omega}{\beta}}_{\tau_m, \tau_{m+2}} \right) \\ &< (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \sum_{\omega \in \mathcal{S}} 2\omega^{\frac{\beta \omega}{\beta}}_{a, b} . \end{aligned}$$

Hence, $N(\gamma, [a, b])$ can be estimated as

$$N(\gamma, [a, b]) < 2 + \frac{2}{\gamma^{\frac{1}{\beta}}} (|\mathcal{S}| + 1)^{\frac{1}{\beta} - 1} \sum_{\omega \in \mathcal{S}} \omega^{\frac{\beta \omega}{\beta}}_{a, b} . \quad (2.8)$$

- For Young equations, one considers $\omega_{s,t}^{(1)} = C_f(t-s)$ with $\beta_1 = 1$ and $\omega_{s,t}^{(2)} = L_g^p \llbracket x \rrbracket_{p,\Pi[s,t]}^p$ with $\beta_2 = \frac{1}{p}$ and $\mathcal{S} = \{\omega^{(1)}, \omega^{(2)}\}$. Then $\beta = \frac{1}{p}$ and (2.8) has the form

$$N(\gamma, [a, b]) < 2 + \frac{2}{\gamma^p} 3^{p-1} \left[C_f^p (b-a)^p + L_g^p \llbracket x \rrbracket_{p,\Pi[a,b]}^p \right]. \quad (2.9)$$

- For rough equations, one consider $\mathcal{S} = \{\omega^{(1)}, \omega^{(2)}, \omega^{(3)}\}$, $\omega_{s,t}^{(1)} = t-s$, $\beta_1 = 1$, $\omega_{s,t}^{(2)} = L_g^p \llbracket x \rrbracket_{p,\Pi[s,t]}^p$, $\beta_2 = \frac{1}{p}$ and $\omega_{s,t}^{(3)} = L_g^q \llbracket \mathbb{X} \rrbracket_{q,\Pi[s,t]}^q$, $\beta_3 = \frac{1}{q} = \frac{2}{p}$. Then $\beta = \frac{1}{p}$ and (2.8) has the form

$$N(\gamma, [a, b]) < 2 + \frac{2}{\gamma^p} 4^{p-1} \left[C_f^p (b-a)^p + L_g^p \llbracket x \rrbracket_{p,\Pi[a,b]}^p + L_g^q \llbracket \mathbb{X} \rrbracket_{q,\Pi[a,b]}^q \right]. \quad (2.10)$$

Rough difference equations

Discrete time set $\Pi = \{t_k\}_{k \in \mathbb{N}}$ defined by

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k, t_{k+1}}(y, \mathbf{x}), \quad (2.11)$$

$$F_{s,t}(y, \mathbf{x}) = \begin{cases} g(y_s)x_{s,t}, & \mathbf{x} = x \\ g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}, & \mathbf{x} = (x, \mathbb{X}) \end{cases} \quad (2.12)$$

$(\mathbf{H}_f)$ $f \in C(\mathbb{R}^d, \mathbb{R}^d)$ of linear growth, i.e.

$$\exists C_f > 0 : \quad \|f(y)\| \leq C_f \|y\| + \|f(0)\|, \quad \forall y \in \mathbb{R}^d; \quad (2.13)$$

$(\mathbf{H}_g)$ either $g \in C_b^2(\mathbb{R}^d, \mathbb{R}^{d \times m})$ such that

$$\|g\|_\infty, C_g := \max \left\{ \|Dg\|_\infty, \|D^2g\|_\infty \right\} < \infty. \quad (2.14)$$

- If $F_{s,t} = g(y_s)x_{s,t}$ then by assigning $L_g := C_g$ one obtains

$$\|\delta F_{s,u,t}\| \leq C_g \|y_{s,u}\| \|x_{u,t}\| \leq L_g \|y\|_{p,\Pi[s,t]} \|x\|_{p,\Pi[s,t]}. \quad (2.15)$$

- If $F_{s,t} = g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}$ where $g \in C_b^2$ or $g(y) = Cy + g(0)$ then a direct computation shows that

$$\begin{aligned} (\delta F)_{s,u,t} &= \left[- \int_0^1 Dg(y_s + \eta y_{s,u})(g(y_s)x_{s,u} + R_{s,u}^y) d\eta \right] x_{u,t} \\ &\quad + Dg(y_s)g(y_s)x_{s,u} \otimes x_{u,t} \\ &\quad + [Dg(y_s)g(y_s) - Dg(y_u)g(y_u)]\mathbb{X}_{u,t}. \end{aligned}$$

From the assumption (H_g) , $Dg(y)g(y)$ is globally Lipschitz continuous with constant L_g^2 given by

$$L_g = \begin{cases} \sqrt{C_g^2 + C_g \|g\|_\infty} & \text{if } g \in C_b^2; \\ C_g & \text{if } g(y) = Cy + g(0). \end{cases}$$

- Define

$$R_{s,t}^y := y_{s,t} - g(y_s)x_{s,t}, \quad s, t \in \Pi.$$

As a result, it is easy to check that for a generic constant $K > 0$

$$\begin{aligned}
\|(\delta F)_{s,u,t}\| &\leq L_g \|R_{s,u}^y\| \|x_{u,t}\| + L_g^2 \|y_{s,u}\| (\|x_{s,u} \otimes x_{u,t}\| + \|\mathbb{X}_{u,t}\|) \\
&\leq L_g \|R_{s,u}^y\| \|x_{u,t}\| \\
&\quad + L_g^2 \|y_{s,u}\| (\|\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t}\| + \|\mathbb{X}_{u,t}\|) \\
&\leq K (L_g \|R_{q,\Pi[s,t]}^y\| \|x_{p,\Pi[s,t]}\| \\
&\quad + L_g^2 \|y_{p,\Pi[s,t]}\| \|\mathbb{X}_{q,\Pi[s,t]}\|). \tag{2.16}
\end{aligned}$$

- It is easy to check that $\left(\|y\|_{p,\Pi[s,t]} \|x\|_{p,\Pi[s,t]} \right)^{\frac{p}{2}}$ for $1 \leq p < 2$, and $(\|R^y\|_{q,\Pi[s,t]} \|x\|_{p,\Pi[s,t]})^{\frac{p}{3}}$ and $(\|y\|_{p,\Pi[s,t]} \|\mathbb{X}\|_{q,\Pi[s,t]})^{\frac{p}{3}}$ for $2 \leq p < 3$, are control functions. Hence from now on, we can write symbolically $\omega_{s,t}^{\lambda_F}(\delta F)$ to indicate the right hand side of (2.15) or (2.16), where $\lambda_F = \frac{2}{p}$ for $1 < p < 2$ and $\lambda_F = \frac{3}{p}$ for $2 \leq p < 3$.

Lemma

Lemma: Let $\{\tau_i\}_{i=\overline{0,m}} \subset \Pi$ be an sub partition of $[a, b]$. The following estimates hold

$$\begin{aligned} \|y\|_{p,\Pi} &\leq m^{\frac{p-1}{p}} \sum_{k=0}^{m-1} \|y\|_{p,\Pi[\tau_k, \tau_{k+1}]} ; \\ \|R^y\|_{q,\Pi} &\leq (2m)^{\frac{q-1}{q}} \left(\sum_{k=0}^{m-1} \|R^y\|_{q,\Pi[\tau_k, \tau_{k+1}]} + C_g m^{\frac{1}{p}} \|x\|_{p,\Pi} \sum_{k=0}^{m-1} \|y\|_{p,\Pi[\tau_k, \tau_{k+1}]} \right). \end{aligned} \quad (2.17)$$

Proof of Lemma: Observe that

$$\begin{aligned} \|(\delta R^y)_{s,u,t}\| &\leq C_g \|y_{s,u}\| \|x_{u,t}\| \\ \Rightarrow \|R^y_{s,t}\| &\leq \|R^y_{s,u}\| + \|R^y_{u,t}\| + C_g \|y_{s,u}\| \|x_{u,t}\|, \quad \forall s \leq u \leq t. \end{aligned} \quad (2.18)$$

Solution estimate

Theorem: Consider the pure rough difference equation

$$z_{t_{k+1}} = z_{t_k} + F_{t_k, t_{k+1}}(z, \mathbf{x}) = z_{t_k} + g(z_{t_k})x_{t_k, t_{k+1}} + Dg(z_{t_k})g(z_{t_k})\mathbb{X}_{t_k, t_{k+1}}.$$

Define $R_{s,t}^z = z_{s,t} - g(z_s)x_{s,t}$. There exists a generic constant K dependent only on p such that the following estimates holds

$$\begin{aligned} \|(z, R^z)\|_{p, \Pi[a,b]} &\leq KN(\gamma, \mathbf{x}, \Pi[a, b])^{\frac{2p-1}{p}} (L_g \|\mathbf{x}\|_{p, \Pi[a,b]} + L_g^2 \|\mathbf{x}\|_{p, \Pi[a,b]}^2); \\ \|\bar{z} - z\|_{\infty, \Pi[a,b]} &\leq e^{3\gamma N^*(\gamma, \mathbf{x}, [a,b])} \|\bar{z}_a - z_a\|. \end{aligned} \quad (2.19)$$

Proof

It follows from (2.16) and the discrete sewing lemma that

$$\begin{aligned}
 \|z_{s,t}\| &\leq \|F_{s,t}\| + \left\| \sum_{\Pi[s,t]} F_{t_k, t_{k+1}} - F_{s,t} \right\| \\
 &\leq L_g \|x_{s,t}\| + L_g^2 \|\mathbb{X}_{s,t}\| \\
 &\quad + K(L_g \|\mathbb{R}^z\|_{q, \Pi[s,t]} \|x\|_{p, \Pi[s,t]} + L_g^2 \|z\|_{p, \Pi[s,t]} \|\mathbb{X}\|_{q, \Pi[s,t]}).
 \end{aligned}$$

As a result,

$$\begin{aligned}
 \|z\|_{p, \Pi[s,t]} &\leq (L_g \|x\|_{p, \Pi[s,t]} + L_g^2 \|x\|_{p, \Pi[s,t]}^2) \\
 &\quad + K \|(z, R^z)\|_{p, \Pi[s,t]} (L_g \|x\|_{p, \Pi[s,t]} + L_g^2 \|x\|_{p, \Pi[s,t]}^2).
 \end{aligned}$$

it follows from the assumptions of f and g that

$$\|(z, R^z)\|_{p, \Pi[s,t]} \leq 2(L_g \|x\|_{p, \Pi[s,t]} + L_g^2 \|x\|_{p, \Pi[s,t]}^2) \left(1 + K \|(z, R^z)\|_{p, \Pi[s,t]}\right). \quad (2.20)$$

This motivates us to construct the greedy sequence of times $\tau_m = \{\tau_m(\gamma, [a, b])\}$ for the discrete time set Π on $[a, b]$ where $\gamma \in (0, 1)$ is a certain positive constant. Observe that whenever $4K(L_g \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]} + L_g^2 \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]}^2) \leq \gamma < 1$ then by taking $\|(z, R^z)\|_{p, \Pi[s, t]}$ from the right hand side to the left hand side of (2.20), one obtains

$$\|(z, R^z)\|_{p, \Pi[\tau_m, \tau_{m+1}]} \leq 4(L_g \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]} + L_g^2 \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]}^2) \leq \frac{\gamma}{K}.$$

On the other hand, whenever $4K(L_g \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]} + L_g^2 \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]}^2) > \gamma$ then τ_m and τ_{m+1} are two consecutive moments in Π , hence

$$\begin{aligned} \|(z, R^z)\|_{p, \Pi[\tau_m, \tau_{m+1}]} &= \|z_{\tau_m, \tau_{m+1}}\| + \|Dg(z_{\tau_m})g(z_{\tau_m})\mathbb{X}_{\tau_m, \tau_{m+1}}\| \\ &\leq 4(L_g \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]} + L_g^2 \|\mathbf{x}\|_{p, \Pi[\tau_m, \tau_{m+1}]}^2). \end{aligned}$$

This together with (2.17) shows that

$$\|(z, R^z)\|_{p, \Pi[a, b]} \leq CN(\gamma, \mathbf{x}, \Pi[a, b])^{\frac{2p-1}{p}} (L_g \|\mathbf{x}\|_{p, \Pi[a, b]} + L_g^2 \|\mathbf{x}\|_{p, \Pi[a, b]}^2)$$

Next, for two solutions $\bar{z}, z$, define $h_t = \bar{z}_t - z_t$ and $R^h = R^{\bar{z}} - R^z$. By similar arguments in [LHD(2022), Proposition 3.3]

$$\begin{aligned} \left\| h, R^h \right\|_{\rho, \Pi[s, t]} &\leq 2K \left\{ L_g \left\| \mathbf{x} \right\|_{\rho, \Pi[s, t]} + L_g^2 \left\| \mathbf{x} \right\|_{\rho, \Pi[s, t]}^2 \right\} \\ &\quad \times \left(\frac{1}{2K} + \left\| \bar{z}, R^{\bar{z}} \right\|_{\rho, \Pi[s, t]} + \left\| z, R^z \right\|_{\rho, \Pi[s, t]} \right) \\ &\quad \times \left(\left\| h_s \right\| + \left\| h, R^h \right\|_{\rho, \Pi[s, t]} \right). \end{aligned} \quad (2.21)$$

and consequently $\left\| h_{\tau_{m+1}} \right\| \leq \left\| h \right\|_{\infty, \Pi[\tau_m, \tau_{m+1}]} \leq (1 + \gamma) \left\| h_{\tau_m} \right\|$ whenever $4K(L_g \left\| \mathbf{x} \right\|_{\rho, \Pi[\tau_m, \tau_{m+1}]} + L_g^2 \left\| \mathbf{x} \right\|_{\rho, \Pi[\tau_m, \tau_{m+1}]}^2) \leq \gamma$. Meanwhile, if $4K(L_g \left\| \mathbf{x} \right\|_{\rho, \Pi[\tau_m, \tau_{m+1}]} + L_g^2 \left\| \mathbf{x} \right\|_{\rho, \Pi[\tau_m, \tau_{m+1}]}^2) > \gamma$ then τ_m and τ_{m+1} are two consecutive moments in Π the sequence $\tau^*(\gamma, \mathbf{x}, [a, b])$ such that $\tau_l^* \leq \tau_m \leq \tau_{m+1} \leq \tau_L^*$

$$\begin{aligned} \left\| h_{\tau_{m+1}} \right\| \leq \left\| h \right\|_{\infty, \Pi[\tau_m, \tau_{m+1}]} &\leq \left(1 + L_g \left\| \mathbf{x}_{\tau_m, \tau_{m+1}} \right\| + L_g^2 \left\| \mathbb{X}_{\tau_m, \tau_{m+1}} \right\| \right) \left\| h_{\tau_m} \right\| \\ &\leq \prod_{k=l}^{L-1} \left(1 + L_g \left\| \mathbf{x}_{\tau_k^*, \tau_{k+1}^*} \right\| + L_g^2 \left\| \mathbb{X}_{\tau_k^*, \tau_{k+1}^*} \right\| \right) \left\| h_{\tau_m} \right\| \\ &\leq e^{(L-l)\gamma} \left\| h_{\tau_m} \right\|. \end{aligned}$$

By induction, $\left\| h \right\|_{\infty, \Pi[a, b]} \leq e^{\gamma(N(\gamma, \mathbf{x}, \Pi[a, b]) + N^*(\gamma, \mathbf{x}, [a, b]))} \left\| h_a \right\| \leq e^{3\gamma N^*(\gamma, \mathbf{x}, [a, b])} \left\| h_a \right\|$ which yields (2.19).

Euler scheme

Theorem [LHD, Kloeden (2021)]: Consider the RDE

$$d\hat{y}_t = f(\hat{y}_t)dt + g(\hat{y}_t)dX_t, \hat{y}_a \in \mathbb{R}^d, t \in [a, b] \quad (2.22)$$

where X is a stochastic process with stationary increment which almost all realizations x are of local Hölder continuous with exponent in $(1/3, 1/2]$ such that x is truly rough.

If in addition to $(\mathbf{H}_f)$, $(\mathbf{H}_g)$, f and D^2g are local Lipschitz, then there exists a generic random variable $\hat{\xi}(\hat{y}_a, x, [a, b])$ such that the following estimate holds

$$\begin{aligned} & \sup_{0 \leq k \leq m} \|\hat{y}(t_k, y_a, x) - y_{t_k}\| \\ & \leq \hat{\xi}(\hat{y}_a, x, [a, b]) (|\Pi[a, b]| + \|x\|_{p, [a, b], |\Pi[a, b]|})^{3-p} (b - a + \|x\|_{p, [a, b]}^p). \end{aligned} \quad (2.23)$$

In particular,

$$\lim_{|\Pi[a, b]| \rightarrow 0} \sup_{0 \leq k \leq m} \|\hat{y}(t_k, \hat{y}_a, x) - y_{t_k}\| = 0. \quad (2.24)$$

Probabilistic setting

Bailleul-Riedel-Scheutzow [2017]. Generation of random dynamical systems and rough flows from rough differential equations.

$p \in (0, 3)$ and a regular discrete time set $\Pi^\Delta = \{k\Delta : k \in \mathbb{Z}\}$, denote by $\mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ the space of all paths $\mathbf{g} : \Pi^\Delta \rightarrow T_1^2(\mathbb{R}^m)$. Then $\mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ is equipped with the compact open topology given by the p -variation norm (2.2), i.e the topology generated by the metric

$$d_p^\Delta(\mathbf{g}, \mathbf{h}) := \sum_{k \geq 1} \frac{1}{2^k} (\|\mathbf{g} - \mathbf{h}\|_{p, \Pi[-k\Delta, k\Delta]} \wedge 1).$$

As a result, it is separable and thus a Polish space.

Let us consider a stochastic process $\bar{\mathbf{X}}$ defined on a probability space $(\bar{\Omega}, \bar{\mathcal{F}}, \bar{\mathbb{P}})$ with realizations in $(\mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m)), \mathcal{F})$. Assume further that $\bar{\mathbf{X}}$ has stationary increments. Assign $\Omega := \mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ and equip it with the Borel σ -algebra $\mathcal{F}$ and let $\mathbb{P}$ be the law of $\bar{\mathbf{X}}$.

Denote by θ the *Wiener-type shift*

$$(\theta_t \omega)_\cdot = \omega_t^{-1} \bullet \omega_{t+\cdot}, \quad \forall t \in \Pi^\Delta, \omega \in \mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m)), \quad (3.1)$$

and define the so-called *diagonal process* $\mathbf{X} : \Pi^\Delta \times \Omega \rightarrow T_1^2(\mathbb{R}^m)$, $\mathbf{X}_t(\omega) = \omega_t$ for all $t \in \Pi^\Delta, \omega \in \Omega$. Due to the stationarity of $\bar{\mathbf{X}}$, it can be proved that θ is invariant under $\mathbb{P}$, then forming a discrete (and thus measurable) dynamical system on $(\Omega, \mathcal{F}, \mathbb{P})$, Theorem 5-(BRSch17). Moreover, $\mathbf{X}$ forms an *p-rough path cocycle*, namely, $\mathbf{X}_\cdot(\omega) \in \mathcal{C}_0^p(\Pi^\Delta, T_1^2(\mathbb{R}^m))$ for every $\omega \in \Omega$, which satisfies the *cocycle relation*:

$$\mathbf{X}_{t+s}(\omega) = \mathbf{X}_s(\omega) \bullet \mathbf{X}_t(\theta_s \omega), \forall \omega \in \Omega, t, s \in \Pi^\Delta,$$

in the sense that $\mathbf{X}_{s,s+t} = \mathbf{X}_t(\theta_s \omega)$ with the increment notation $\mathbf{X}_{s,s+t} := \mathbf{X}_s^{-1} \bullet \mathbf{X}_{s+t}$. In particular, the Wiener shift (3.1) implies that

$$\|\mathbf{x}(\theta_h \omega)\|_{p, \Pi[s, t]} = \|\mathbf{x}(\omega)\|_{p, \Pi[s+h, t+h]}, \quad \forall s, t, h \in \Pi^\Delta. \quad (3.2)$$

Discrete RDS

Define the discrete mapping $H^\Delta(\mathbf{x})y = y + f(y)\Delta + F_{0,\Delta}(y, \mathbf{x})$. Similar to Duc-Kloeden, we can easily prove that the discrete Euler scheme (2.11) generates a discrete random dynamical system $\varphi^\Delta : \Pi^\Delta \times \Omega \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ over $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ such that

$$\begin{aligned}\varphi^\Delta(0, \omega)y_0 &= y_0, \\ \varphi^\Delta(k\Delta, \omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega) \circ \dots \circ H^\Delta(\omega)y_0, \quad \forall k \geq 1.\end{aligned}\tag{3.3}$$

Throughout this paper, we assume that θ is an ergodic dynamical system.

Ergodicity of θ

- Garrido-Atienza, Schmalfuss (2011): the Wiener shift $\eta_t x = x_{t+} - x_t$ is ergodic on $\Omega' = C^{0,\beta}(\mathbb{R}, \mathbb{R}^m)$ w.r.t. $\mathbb{P}_H$.
- Friz-Hairer (2014, Theorem 10.4): there exists a full measure subset $\Omega_1 \subset \Omega'$ such that $\omega = (1, x, \mathbb{X}) \in \Omega$ for any $x \in \Omega_1$. Moreover, by Friz-Victoir (2010, Theorem 15.42, 45), one can choose this full measure subset Ω_1 such that it satisfies the piecewise linear approximations (mollifier approximation).
- Then consider the natural lift $\mathcal{S}$ on smooth paths

$$\mathcal{S}(x)_{s,t} = (1, x_{s,t}, \int_s^t x_{s,r} dx_r), \quad (3.4)$$

which can be extended to Ω_1 such that $\mathcal{S} : \Omega_1 \rightarrow \Omega$. By Bailleul, Riedel, Scheutzow (2017), there exists a metric dynamical system $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}}, \theta)$ such that $\hat{\Omega} \subset \Omega$ and $\hat{\mathbb{P}} = \mathbb{P}^H \circ \mathcal{S}^{-1}$. Furthermore,

$$\theta_t \mathcal{S}(x) = \lim_{n \rightarrow \infty} \hat{\theta} \mathcal{S}(x^{(n)}) = \lim_{n \rightarrow \infty} \mathcal{S}(\eta_t x^{(n)}) = \mathcal{S}(\eta_t x). \quad (3.5)$$

- Since η is ergodic, by Garrido-Atienza, Schmalfuss (2011, Lemma 3), θ is also ergodic.
- From now on, we assume additionally that $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ is ergodic.

Pullback attractors

Roughly speaking, an invariant random compact set $\mathcal{A} \in \mathcal{D}$ is called a *pullback attractor* in $\mathcal{D}$, if $\mathcal{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the pullback sense, i.e.

$$\lim_{t \rightarrow \infty} d_H(\varphi(t, \theta_{-t}\omega) \hat{D}(\theta_{-t}\omega) | \mathcal{A}(\omega)) = 0, \quad (3.6)$$

where $d_H(\cdot | \cdot)$ is the Hausdorff semi-distance, i.e.

$d_H(D | A) := \sup_{d \in D} \inf_{a \in A} \|d - a\|$. A random set $\mathcal{B} \in \mathcal{D}$ is called *pullback absorbing* in the universe $\mathcal{D}$ if $\mathcal{B}$ absorbs all closed random sets in $\mathcal{D}$, i.e. for any closed random set $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega) \hat{D}(\theta_{-t}\omega) \subset \mathcal{B}(\omega), \text{ for all } t \geq t_0. \quad (3.7)$$

$$\mathcal{A}(\omega) = \bigcap_{t \geq 0} \overline{\bigcup_{s \geq t} \varphi(s, \theta_{-s}\omega) \mathcal{B}(\theta_{-s}\omega)}. \quad (3.8)$$

Existence of continuous attractors

Now we assume further that f is dissipative: there exist $c, d > 0$ so that for all $y \in \mathbb{R}^d$,

$$\langle y, f(y) \rangle \leq c - d\|y\|^2. \quad (3.9)$$

In addition, assume the p -variation norm of x is of finite moment for all order $k \in \mathbb{N}$, i.e.

$$\mathbb{E} \|\mathbf{x}(\cdot)\|_{p, \Pi[a, b]}^k < \infty, \quad \forall k \in \mathbb{N}, \quad \forall 0 < a < b, \quad a, b \in \Pi. \quad (3.10)$$

Theorem [LHD(2022)]: Assume $(\mathbf{H}'_f), (\mathbf{H}^b_g), (\mathbf{H}_X)$, then there exists a random pullback attractor $\mathcal{A}$ such that $|\mathcal{A}(\cdot)| \in \mathcal{L}^\rho$ for any $\rho \geq 1$. If in addition $(\mathbf{H}_{\mathcal{A}})$ and f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that $\mathcal{A} \rightarrow \mathcal{A}$ a.s. (w.r.t. the Hausdorff semi-distance) as $C_g \rightarrow 0$, both in the almost sure and in $\mathcal{L}^\rho$ senses. Moreover, if f is strictly dissipative then $\mathcal{A}$ is a singleton provided that C_g is sufficiently small.

Compactness arguments

Lemma: For any $\lambda > 0$ small enough, there exist constants $\delta_\lambda, C_\lambda > 0$ such that for any solution μ_t of the ODE lying in the global attractor $\mathcal{A}$, the following estimates hold

$$g \in C_b^3 : \|y_t\| \leq \|y_0\| e^{-\delta_\lambda t} + C_\lambda N\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [0, t]\right);$$

To estimate $\|z_t\|$, we rewrite (1.11) as

$$\dot{z}_t = (Id + \psi_t)f(z_t + \gamma_t). \quad (3.11)$$

First, we are going to prove that there exists constants $\bar{C}_\lambda, \delta_\lambda > 0$ such that

$$\frac{d}{dt} \|z_t\|^2 \leq \bar{C}_\lambda - \delta_\lambda \|z_t\|^2, \quad \forall t \in [0, \tau]. \quad (3.12)$$

Hence by Gronwall lemma and Cauchy inequality, one obtains

$$\|z_\tau\| \leq \|z_0\| e^{-\delta_\lambda \tau} + \frac{\bar{C}_\lambda}{\delta_\lambda} \Rightarrow \|y_\tau\| \leq \|z_\tau\| + \|\eta\| \leq \|y_0\| e^{-\delta_\lambda \tau} + \frac{\bar{C}_\lambda}{\delta_\lambda} + \frac{\lambda}{2}.$$

By constructing a greedy sequence of stopping times and using induction, one can easily show that

$$\|y_{\tau_i}\| \leq \|y_0\| \exp\left\{-\delta_\lambda \tau_i\right\} + iC_\lambda, \quad \forall i \in \mathbb{N}.$$

Existence of discrete attractors

Theorem [ND. Cong-LHD-PT. Hong(2022)] Assume $(\mathbf{H}_f)$, $(\mathbf{H}_g)$ and the dissipativity condition (3.9) and condition (3.10). Consider system (2.11) with the regular time set Π^Δ , where $\Delta \in (0, 1)$ satisfied the inequality

$$0 < \Delta < 1 \wedge \frac{d}{2C_f^2} \wedge \frac{1}{2d}. \quad (3.13)$$

Then the generated discrete random dynamical system φ^Δ of (2.11) admits a random pullback attractor $\mathcal{A}^\Delta$.

Proof

We would like to compare the solution of the discrete system

$$\bar{y}_{t_{k+1}} = \bar{y}_{t_k} + f(\bar{y}_{t_k})(t_{k+1} - t_k)$$

and the solution y of the equation (2.11) that starts at the same initial condition $\bar{y}_0 = y_0$.

Proposition: Assume that f is Lipschitz continuous, i.e.

$\|f(\bar{y}) - f(y)\| \leq C_f \|\bar{y} - y\|$. Then

$$\|\bar{y} - y\|_{\infty, \Pi[0, T]} \leq \left(\omega(\Pi, \mathbf{x})_{0, T} \|y_0\|^\beta + L_g \|\mathbf{x}\|_{\rho, \Pi[0, T]} \right) e^{3\gamma N^*(\gamma, \mathbf{x}, [0, T])}, \quad (3.14)$$

where $\omega(\Pi, \mathbf{x})_{s, t} = C(L_g \|\mathbf{x}\|_{\rho, \Pi[s, t]} + L_g^2 \|\mathbf{x}\|_{\rho, \Pi[s, t]}^2)(t - s)^\beta$ is a discrete control for $\beta \in (1 - \alpha, 1)$.

Indeed, define $h_t = \bar{y}_t - y_t$. Then $h_0 = 0$ and

$$\begin{aligned} \|h_{s, t}\| &\leq C_f \|h\|_{\infty, \Pi[s, t]}(t - s) + L_g \|\mathbf{x}\|_{\rho, \Pi[s, t]} + L_g^2 \|\mathbf{x}\|_{\rho, \Pi[s, t]}^2 \\ &\quad + \left\| \sum_k F_{t_k, t_{k+1}}(h + \bar{y}, \mathbf{x}) - F_{s, t}(h + \bar{y}, \mathbf{x}) \right\|. \end{aligned}$$

A direct computation shows that

$$\begin{aligned}
\|\delta F_{s,u,t}\| &\leq \|g(\bar{y}_u) - g(\bar{y}_s) - Dg(\bar{y}_s)g(\bar{y}_s)x_{s,u}\| \|x_{u,t}\| \\
&\quad + \|Dg(\bar{y}_u)g(\bar{y}_u) - Dg(\bar{y}_s)g(\bar{y}_s)\| \|\mathbb{X}_{u,t}\| \\
&\leq 2(L_g \|\mathbf{x}\|_{\rho, \Pi[s,t]} \left\| R^h \right\|_{q, \Pi[s,t]} + L_g^2 \|\mathbf{x}\|_{\rho, \Pi[s,t]}^2 \|h\|_{\rho, \Pi[s,t]}) \\
&\quad + 2(L_g \|\mathbf{x}\|_{\rho, \Pi[s,t]} + L_g^2 \|\mathbf{x}\|_{\rho, \Pi[s,t]}^2) \|y\|_{1-\text{var}, [s,t]}^\beta + L_g^3 \|\mathbf{x}\|_{\rho, \Pi[s,t]}^3.
\end{aligned}$$

Then by similar arguments, one can show that

$$\begin{aligned}
\left\| h, R^h \right\|_{\rho, \Pi[s,t]} &\leq 2K \left\{ L_g \|\mathbf{x}\|_{\rho, \Pi[s,t]} + L_g^2 \|\mathbf{x}\|_{\rho, \Pi[s,t]}^2 \right\} \\
&\quad \times \left(\omega(\Pi, \mathbf{x})_{s,t} \|y_0\|^\beta + L_g \|\mathbf{x}\|_{\rho, \Pi[s,t]} + \left\| h, R^h \right\|_{\rho, \Pi[s,t]} \right).
\end{aligned}$$

Thus one concludes that

$$\begin{aligned}
\|h\|_{\infty, \Pi[0,1]} &\leq e^{3\gamma N^*(\gamma, \mathbf{x}, [0,1])} (\|h_0\| + \omega(\Pi, \mathbf{x})_{0,1} \|y_0\|^\beta + L_g \|\mathbf{x}\|_{\rho, \Pi[0,1]}) \\
&\leq \left(\omega(\Pi, \mathbf{x})_{0,1} \|y_0\|^\beta + L_g \|\mathbf{x}\|_{\rho, \Pi[0,1]} \right) e^{3\gamma N^*(\gamma, \mathbf{x}, [0,1])}.
\end{aligned}$$

This proves (3.14)

Semi-continuity of attractors

The proof is then a direct consequence of the two estimates

$$\begin{aligned}
 \|\bar{y}_{k+1}\|^2 &\leq (1 - 2d\Delta + 2C_f^2\Delta^2)\|\bar{y}_k\|^2 + 2(c\Delta + 2C_f^2\Delta^2) \quad \text{so that} \\
 \|\bar{y}_1\|^2 &\leq e^{-d}\|\bar{y}_0\|^2 + 2(c\Delta + C_f^2\Delta^2)\frac{1 - e^{-d}}{1 - e^{-d\Delta}} \\
 &\leq e^{-d}\|\bar{y}_0\|^2 + 4(c + C_f^2\Delta)\frac{1 - e^{-d}}{d}
 \end{aligned}$$

and (by Jensen and Young inequalities)

$$\begin{aligned}
 \|y_1\|^2 &\leq (\|\bar{y}_1\| + \xi_1(\mathbf{x}, \Delta)\|y_0\|^\beta + \xi_2(\mathbf{x}, \Delta))^2 \\
 &\leq (1 + \varepsilon)\|\bar{y}_1\|^2 + \varepsilon\|y_0\|^2 + \bar{\xi}(\mathbf{x}, \Delta) \\
 &\leq \left[(1 + \varepsilon)e^{-d} + \varepsilon\right]\|y_0\|^2 + \bar{\xi}(\mathbf{x}, \Delta).
 \end{aligned}$$

Theorem: Under assumptions $(\mathbf{H}_f)$, $(\mathbf{H}_g)$ and for all Δ small enough, the attractor $\mathcal{A}^\Delta$ and the absorbing set $\mathcal{B}^\Delta$ of (2.11) is uniformly bounded in Δ a.s. As a result, for L_g small enough, the numeric attractor $\mathcal{A}^\Delta$ converges to the attractor $\mathcal{A}$ in the Hausdorff semi-distance, i.e. $d_H(\mathcal{A}^\Delta|\mathcal{A}) \rightarrow 0$ as $\Delta \rightarrow 0$, a.s.

Fitzhugh-Nagumo neuro-system

The deterministic FHN neuron model is

$$\begin{cases} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon(v_t - \mu w_t + J) \end{cases} \quad (4.1)$$

It is important to note that although our vector field f is dissipative, it does not satisfy the linear growth in the perpendicular direction

$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y \neq 0 \quad (4.2)$$

for a certain constant C_f , hence one can not apply directly the arguments in [?] and [?] for the standard Lyapunov function $V(y) = \|y\|^2$ to prove the existence and uniqueness theorem. Instead, we introduce a different Lyapunov function

$$V(y) := \left(1 + v^4 + Bw^2\right)^{\frac{1}{4}}, \quad (4.3)$$

where $B > 0$ can be chosen later. Observe that

$$\nabla V(z) = \frac{(4v^3, 2Bw)}{4\left(1 + v^4 + Bw^2\right)^{\frac{3}{4}}},$$

In other words, there exists a constant $L_V > 0$ such that

$$\|\nabla V\| \leq L_V. \quad (4.4)$$

Theorem: Assign $B := \frac{6}{\varepsilon\mu}$. Under the assumptions $(\mathbf{H}_g^b)$, $(\mathbf{H}_X)$, there exists a solution of (2.22) on any interval $[0, T]$. Moreover, there exist constants $\delta > 0$ such that for any $\lambda \in (0, 1)$ small enough, there exist a constant $K_\lambda > 0$ such that the following estimate holds for all

$$V(y_t) - K_\lambda \leq e^{-\delta t} \left[V(y_0) - K_\lambda \right] + L_V \lambda \int_0^t e^{-\delta s} dN^-\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [t-s, t]\right), \quad \forall t \in [0, T] \quad (4.5)$$

where the integral in the right hand side of (4.5) is well-defined as a Riemann-Stieltjes integral driven by an increasing path $N^-(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [t - \cdot, t])$ on $[0, t]$.

Corollary: There exist constants $\delta > 0$ such that for any $\lambda \in (0, 1)$ small enough, there exist a constant $K_\lambda > 0$ such that the following estimate holds for all

$$V(y_t) - K_\lambda \leq e^{-\delta t} \left[V(y_0) - K_\lambda \right] + H(\|\mathbf{x}\|_{p, [0, t]}), \quad \forall t \in [0, T] \quad (4.6)$$

where

$$H(\xi) = L_V (16C_p C_g)^p \lambda^{1-p} \xi^p + 8L_V C_p C_g \xi. \quad (4.7)$$

Open problem: Numerical attractor for rough FHN system.